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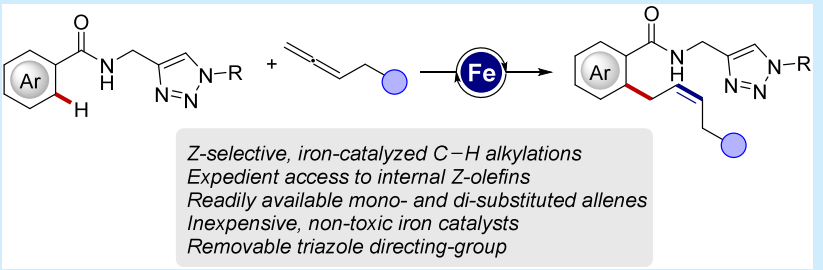
Z-Selective Iron-Catalyzed C−H Alkylations with Allenes: Access to Stereodefined Olefins

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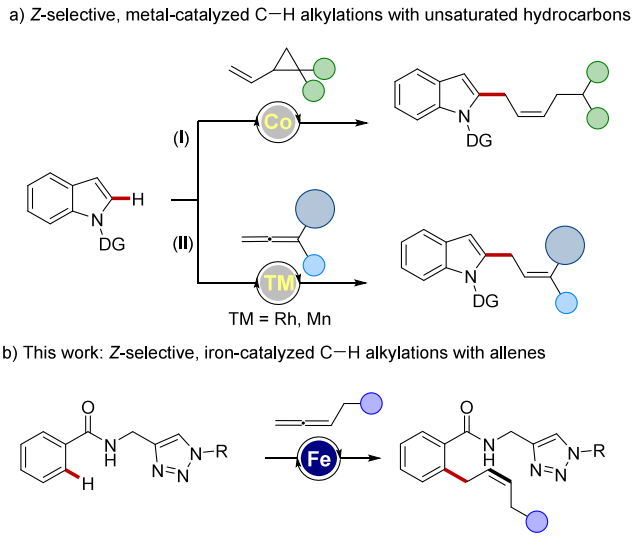
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ABSTRACT: We report an unprecedented iron-catalyzed C−H alkylation of widespread benzamides using readily available allenes. The method, based on cheap, nontoxic, and earth abundant iron catalysts, exploits the assistance of removable triazole directing groups to yield a wide array of *Z* internal olefins, even at room temperature, with excellent stereocontrol, good yields, and broad functional group tolerance.

Olefins represent one of the most important scaffolds in organic chemistry due to their use in the flavor and fragrance industries, material sciences, and naturally occurring molecules with important biological applications.[1] One of the primary challenges associated with their synthesis is the control of the stereochemical configuration (*E* or *Z*), which often influences their inherent chemical and physical properties.[2] While synthesis routes for *E*-olefins are well established, accessing Z-olefins remains a challenging goal.[3] In the last years, transition-metal-catalyzed C−H activations have emerged as a powerful platform for step- and atom-economical molecular transformations.[4] In this context, value-added olefins can be synthesized by means of C−H activations using unsaturated hydrocarbons as reaction patterns.[5] Despite these advances, C−H activation methods to access *Z*-olefins continue to be scarce and are limited to the functionalization of heterocyclic indole substrates. Cobalt catalysts have been exploited for C−H/C−C activations of vinylcyclopropanes for the synthesis of *Z*-olefins (Scheme 1a, **I**).[6] Further, rhodium- or manganese-catalyzed C−H allylations with tailored-designed allenes were recently reported as an effective approach for producing alkenes with high *Z*-selectivities (Scheme 1a, **II**).[7] In this scenario, approaches to *Z*-olefins by chelation-assisted C−H activation of ubiquitous benzamides remain unexplored. Among transition metal catalysts, iron represents the most attractive in terms of its earth abundance, cost efficiency, and low toxicity,[8] and it has recently witnessed significant momentum toward efficient C−H activations.[9]

Scheme 1. *Z*-Selective C−H Alkylations with Unsaturated Hydrocarbons



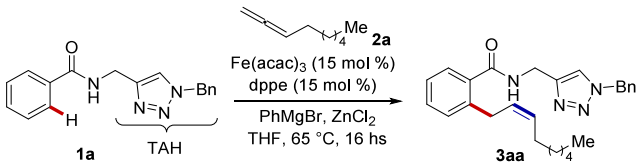
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Intrigued by our recent findings on iron-catalyzed C−H activations by triazole assistance,[10] we wondered whether readily available allenes[11] could be employed to access *Z*-stereoisomers through the geometrical restrictions offered by the triazole chelation[12] (Scheme 1, b). We commenced our studies by probing triazolyl (TAH)-derived[10j] benzamide **1a** in various reaction conditions using monosubstituted allene **2a**. Noteworthily, iron catalysis enabled the envisioned C−H alkylation, leading to the synthesis of **3aa** with exclusive *Z*-selectivity (Table 1).

Table 1. Variation of Reaction Conditions[a]

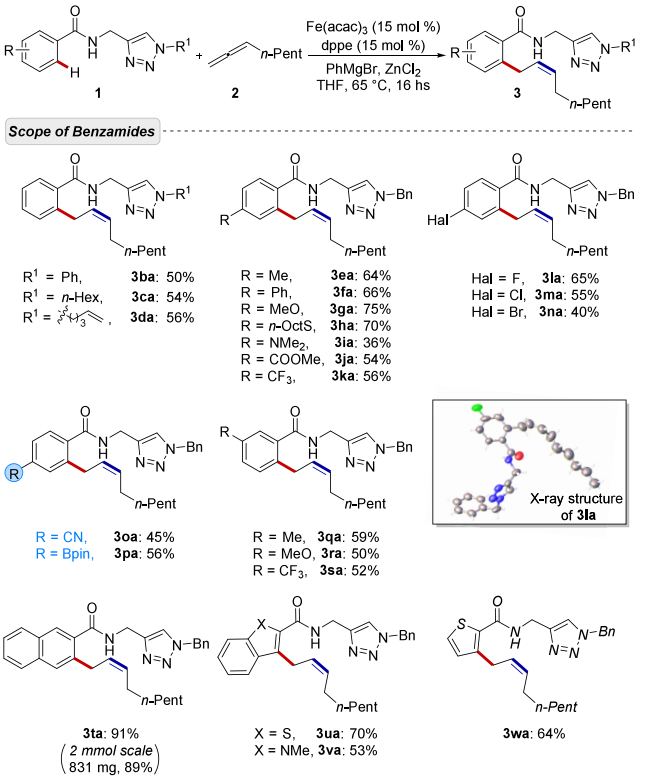


Entry[a]	Variation of reaction conditions	3aa [%][b]
1	none	80 (77)
2	$FeCl_2$ instead of $Fe(acac)_3$	72
3	$CoCl_2$ or $Ni(acac)_2$ or $Pd(OAc)_2$ instead of $Fe(acac)_3$	--
4	2,2′-bipy instead of dppe	--
5	dppen instead of dppe	63 (58)
6	dppbz instead of dppe	11
7	$TMSCH_2MgBr$ instead of PhMgBr	--
8	*i*-PrMgBr instead of PhMgBr	20
9	$ZnCl_2$TMEDA	--
10	No $ZnCl_2$	--
11	25 °C	(70)

[a]Reaction conditions: **1a** (0.2 mmol), **2a** (0.4 mmol), [Fe] (0.03 mmol), ligand (0.03 mmol), RMgBr (0.6 mmol), ZnX_2 (0.4 mmol), THF (0.2 mL). [b]Yields determined using 1,3,5-trimethoxybenzene as the internal standard. In parentheses, isolated yields. dppe = 1,2-bis(diphenylphosphino)ethane; dppbz = 1,2-bis(diphenylphosphino)-benzene; dppen = *cis*-1,2-bis(diphenylphosphino)ethene.

The optimal reaction conditions devised for the *Z*-selective C−H alkylation required a bidentate diphosphine ligand such as dppe, $ZnCl_2$ (2.0 equiv), and PhMgBr (3.0 equiv) as the base. Other iron precatalysts such as $FeCl_2$ were found to be suitable for the reaction (entry 2), while broadly employed transition metals such as cobalt, nickel, or palladium did not provide any transformation under otherwise identical reaction conditions (entry 3). This outcome reflected the unique nature of the iron catalytic regime. Among the ligands tested, 2,2′-bipyridine (entry 4) failed to promote the C−H alkylation, while the backbone of the diphosphine ligands significantly influenced the outcome. In fact, the use of more rigid scaffolds such as dppen and dppbz led to product **3aa** in lower yields (entries 5, 6). The use of alkyl magnesium reagents was detrimental to the catalysis, in contrast to previously reported iron-catalyzed C−H annulations (entries 7, 8).[10,10d,e,h] Zinc salts were crucial, as well. Hence, when $ZnCl_2$ was employed as a TMEDA adduct, a transformation was not observed (entry 9). Likewise, its absence was detrimental to the iron-catalyzed C−H alkylation with allenes, likely due to less effective transmetalation (entry 10).[13] The catalytic transformation was also effective under exceedingly mild conditions at 25 °C. In this latter case, the desired product **3aa** was isolated in good yields of 70% (entry 11). It is important to note that the use of the most popular 8-AQ directing group was not possible for this type of transformation.[14] Despite numerous efforts (see Supporting Information for more details), we were not able to isolate any C−H alkylation products here. Under the optimized catalytic conditions, we next probed the versatility of the *Z*-selective, iron-catalyzed C−H alkylation protocol in the presence of benzamide derivatives (Scheme 2). Differently

Scheme 2. Substrate Scope for Benzamides 1

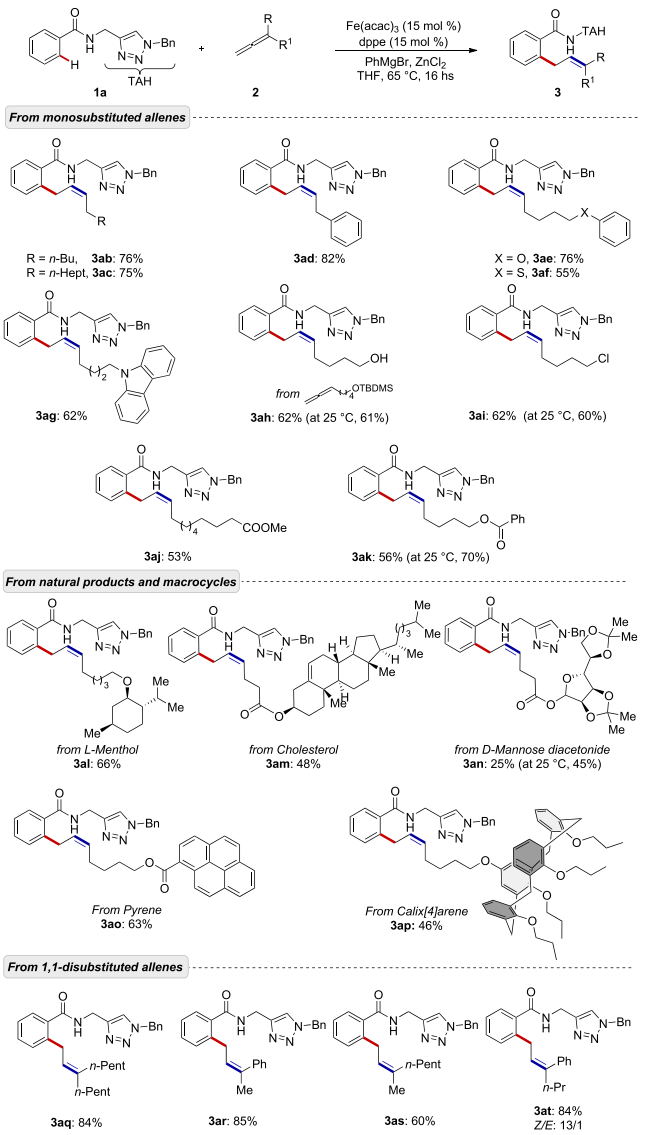


substituted methylene-tethered triazoles **1b−d** were all applicable, although they delivered the corresponding products **3ba−da** in only moderate yields. Various arenes, functionalized with both electron-rich and electron-withdrawing groups at the *para* and *meta* positions, were all generally well-applicable. Also, challenging nitrile and boronate[15] derivatives were well tolerated under the catalytic conditions (**3oa−pa**). Finally, heterocyclic derivatives were selectively converted into desired products **3ua−wa**.

Subsequently, a set of diversely decorated allenes **2** was utilized for iron-catalyzed C−H alkylations (Scheme 3). Alkyl- and aryl-substituted allenes **2** were thus efficiently converted into the corresponding products **3ab−ad**. Allenes attached to (thio)ethers and carbazole moieties were also well-tolerated (**3ae−ag**). A silyl-allenyl ether directly yielded the corresponding unprotected alcohol **3ah** upon acidic workup of the catalytic reaction. The iron catalysis displayed a remarkable functional group tolerance since compounds **3ai−ak**, functionalized with alkyl chlorides and esters, were all produced in good yields. Prompted by these findings, we probed the iron-catalyzed method in the presence of monosubstituted allenes decorated with complex synthetic scaffolds. For instance, L-menthol- and cholesterol-functionalized benzamides **3al**,**am** were delivered in synthetically useful yields as well as D-mannose diacetonide derivative **3an**. Furthermore, the applicability of the C−H alkylation method was demonstrated in the presence of pyrene and calix[4]arenes (**3ao**,**ap**).[16] The

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Org. Lett. 2025, 27, 7950−7955

Scheme 3. Substrate Scope for Allenes 2



reactivity of 1,1-disubstituted allenes was further tested. A dialkyl derivatives such as **2q** was reacted under standard conditions, leading to product **3aq** in high yields. The outstanding *Z*-selectivity was retained for asymmetrical disubstituted allenes. Particularly, when allenes **2r**−**t** were submitted to iron catalysis, the corresponding products **3ar**−**at** were formed with complete to high stereoselectivity, in very good yields.

To gain insights into the origin of the experimentally observed *Z*-selectivity, DFT calculations were carried out for the migratory insertion elementary step at the PW6B95-D4/def2-TZVP+SMD(THF)//TPSS-D3(BJ)/def2-SVP level of theory (Figure 1) (see SI for more details). Given the electronic nature of the iron(II) complexes, high-spin (quintet), low-spin (singlet), and an intermediate triplet state were taken into consideration for both *Z*- and *E*-selective pathways. For the latter two scenarios, a spin-crossover is at play, from a triplet (**I-1**) to a singlet surface upon allene coordination (**I-2**). In the latter, the *Z*-pathway (**I-2Z**) was shown to be energetically favorable over the *E*-pathway (**I-2E**) by 5.9 kcal mol^{-1}. Such an accentuated stabilization is associated with less steric hindrance upon allene coordination.

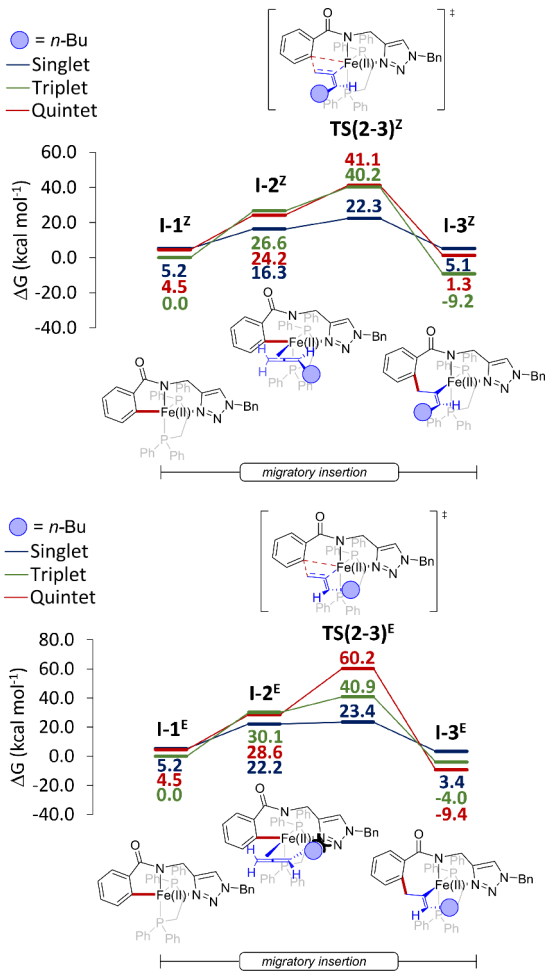


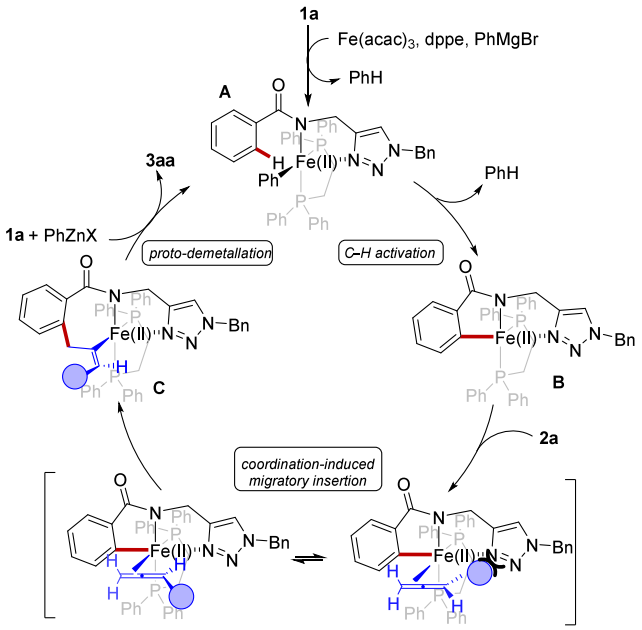
Figure 1. Computed relative Gibbs free energies (ΔG) in kcal mol^{-1} for the allene coordination-induced migratory insertion to access both *Z*- and *E*-selectivity pathways (top and bottom, respectively) at the PW6B95-D4/def2-TZVP+SMD(THF)//TPSS-D3(BJ)/def2-SVP level of theory.

Afterward, coordination-induced migratory insertion proceeds through transition state **TS(2−3)**, where the *Z*-pathway (**TS(2−3)Z**) was calculated to be favored over the *E*-pathway (**TS(2−3)E**) by an overall barrier of 1.1 kcal mol^{-1}. Such results provide strong support for the experimentally obtained *Z*-selectivity.

Based on several experimental evidence (see SI for more details) and our detailed computational analysis,[17] we propose a catalytic cycle initiated by a facile C−H activation that proceeds via a ligand-to-ligand hydrogen transfer (LLHT)[18] leading to **B**, as depicted in Scheme 4. Subsequently, a stereodetermining, coordination-induced migratory insertion of the allene occurs, forming the key cyclometalated species **C**. The origin of the high stereoselectivity toward the formation of the *Z*-alkenes could be attributed to repulsive steric interactions in the migratory allene insertion. Finally, proto-demetalation delivers product **3aa**, regenerating the active iron species **A** (Scheme 4). The observed high levels of chemoselectivity toward proto-demetalation over annulation are likely due to the nature of the allene substitution pattern, contrasting with our previous findings on annulations. In the latter the leaving groups on the allene were shown to aid the C−N forming reductive elimination from intermediate **C**[10a,e]

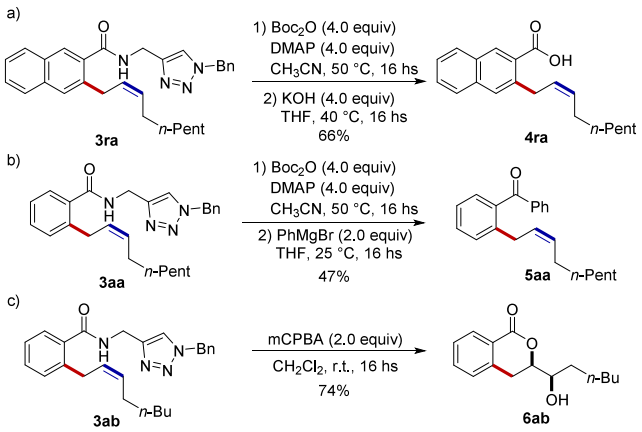
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Scheme 4. Plausible Reaction Mechanism



The synthetic utility of the iron-catalyzed C−H alkylation protocol was finally demonstrated through several orthogonal diversifications of the triazole directing group. As in the example, compound **3ra** was transformed into the corresponding carboxylic acid **4ra** by a first protection of the secondary amide as a *tert*-butyloxycarbamate and subsequent basic hydrolysis (Scheme 5, a). Noteworthily, **4ra** retained the *Z*-

Scheme 5. Removal of TAH and Late-Stage Modifications



geometry of the olefinic fragment. The functionalization of **3aa** as a carbamate was further exploited to directly convert the amide into the corresponding ketone **5aa** by simple reaction with PhMgBr (Scheme 5, b).[19] Finally, the use of *m*-CPBA enabled an unprecedented oxidative deprotection strategy for **3ab**, leading to the synthetically versatile 3,4-dihydrocoumarin derivative **6ab** (Scheme 5, c).[20]

In conclusion, we reported on a strategy for the stereoselective synthesis of internal *Z*-olefins by means of iron-catalyzed C−H alkylations with allenes. The iron catalysis exploited the features of bioisosteric triazoles to deliver a family of benzamides with ample scope and functional group tolerance, even in the presence of challenging boronate esters or nitriles. The triazole could be removed in a traceless fashion, leading to biorelevant compounds.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c02274.

General remarks and materials, mechanistic studies, X-ray crystallography, and copies of NMR spectra (PDF)

Accession Codes

Deposition Numbers 2427040−2427042 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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